



SABITA ROY

Ph. D awarded from Jadavpur University

📍 Kolkata West Bengal 700032 📞 8336044606, 8902649623 ✉ Sabita21.roy@gmail.com

Professional Summary

Innovative Assistant Professor bringing Proven success in implementing technology-based curriculum delivery and assessment tools. Passionate about fostering academic development and success for every student.

Skills

- Student records management
- Lesson planning
- Discussion facilitation
- Presentations
- Online learning tools
- Time management
- Attention to detail
- Microsoft Office
- ERDAS
- Arc GIS
- PCM Geometica
- SPSS
- CROPWAT
- CLIMEWAT
- QGIS
- IDRISI
- SWAT
- SNAP

Education

- Ph.D. in Geography** **JU, Kolkata**
Jadavpur University, School of Oceanographic Studies 2022
Thesis Title: Assessment of Drought Dynamics in Purulia District, West Bengal for appropriate adaptation to climate change impacts.
Supervisor: Professor Sugata Hazra
- Post Graduate Diploma in Remote Sensing & GIS** **JU, Kolkata**
Jadavpur University, School of Oceanographic Studies 2015
Grade A (81%)
- MA in Geography** **VU, Medinipur, WB**
Vidyasagar University 2013
First Class (60.7%)
- BA in Geography Honours** **BU, Burdwan, WB**
Burdwan University 2011
Second Class (56.5%)

	H.S First Class (76.2%) Madyamik First Class (79.75%) Qualifications: UGC NET Geography in June 2015 (67.7%)	WBCHSE 2008 WBBSE 2006
Work Experience	Assistant Professor <i>K K University- Nalanda, Bihar</i> • Built strong rapport with students through class discussions and academic advisement. • Evaluated student performance through comprehensive assessments, identifying areas of improvement and providing targeted feedback for growth • Prepare syllabi, curricula, reading materials, tests, assignments and projects • Keep thorough records of over 200 student scores and attendance • Grade exams and paper, giving detailed feedback • Teach six courses in Geography to 90 students at both graduate and post graduate levels.	06/2024 - Current
Research Experience	Climate Change Expert <i>NETZ Partnerschaft für Entwicklung und Gerechtigkeit e.V., DRCSC</i> • Study on the causes of vulnerability and conflict dynamics related to climate change. • Study on existing local adaptive capacities and vulnerabilities of affected communities as well as climate-related conflict dynamics in the Tribal and Schedule Cast Communities in West Bengal (Birbhum) • Developed questionnaires for household-level surveys, and checklists for key informant interviews and focus group discussions • Collect data/information from the selected stakeholders/locations/communities, and analyze the data/information collected from the field • Analyzed climatic data, drought and socio-economic conditions and other variables using primary, secondary and remote sensing data.	06/2023 - 09/2023

Senior Research Fellow

NABARD and Jadavpur University

02/2022 - 05/2023

- Developed and implemented an SDSS model for sustainable water resource management in drought drought-affected RLZ region of West Bengal (Purulia and Bankura) and successfully achieved the project's scheduled goal.
- Combined the traditional knowledge, science and modern technologies for area-specific adaptation strategies and implemented this model at the farm level with excellent results.
- Diversified farmers' income and improved food security by implementing supply side and demand side management as well as helping to mitigate the drought impacts.

Research Assistant

09/2022 -11/2022

NACO and Tata Institute of Social Sciences

- Systematic review and prepared review papers
- Prepared questionnaires for both qualitative and quantitative studies.
- Conducted interviews for field investigators

Junior Research Fellow

NABARD and Jadavpur University

11/2015 - 02/ 2017

- Analysis of the geomorphology (High-Resolution Contour Map, Slop, Lineament, Drainage Density, Micro Watershed Delineation,
- Spatio-temporal drought was analyzed using high-resolution satellite data at the micro-level based on precipitation, vegetation, temperature & soil moisture indices.
- Analyzed the impact of drought on socio-economic conditions and Migration.

Publications

- ◆ **Roy S**, Hazra S, Chanda A (2023) Changing characteristics of meteorological drought and its impact on monsoon-rice production in sub-humid red and laterite zone of West Bengal, India. Theoretical and Applied Climatology.151(3):1419-33. ISSN 1434-4483
- ◆ **Roy S**, Soren R, Hazra S (2023) Benefits of Reviving Indigenous Water Conservation Practices for Drought Resilience in Red and Lateritic Zones

of West Bengal, India. In Indigenous and Local Water Knowledge, Values and Practices (pp. 175-197). Singapore: Springer Nature Singapore

- ◆ **Roy S**, Hazra S, Chanda A (2022) Assessment of wet season agricultural droughts using monthly MODIS and SAR data in the red and lateritic zone of West Bengal, India. Spatial Information Research. <https://doi.org/10.1007/s41324-022-00485-y> ISSN 2366-3294
- ◆ **Roy S**, Hazra S, Chanda A, Das S (2022) Land suitability analysis using AHP-based multi- criteria decision model for sustainable agriculture in red and lateritic zones of West Bengal, India. Journal of Earth System Science 131, 201 (2022). <https://doi.org/10.1007/s12040-022-01941-x>
- ◆ **Roy S**, Hazra S, Chanda A (2022) Identifying Rainwater Harvesting Structure Sites using MCDM based GIS approach: A Mitigation Measure for Drought in sub-humid red and lateritic zones of West Bengal, India. Arabian Journal of Geoscience. DOI: 10.1007/s12517-022-10077-7 ISSN 1866-7538
- ◆ **Roy S**, Hazra S, Chanda A, Das S (2020) Assessment of groundwater potential zones using multi-criteria decision-making technique: a micro-level case study from red and lateritic zone (RLZ) of West Bengal, India. Sustainable Water Resources Management 6(1):4. ISSN 23635049
- ◆ **Roy S**, Hazra S (2020) Monthly drought assessment and early warning for drought in Purulia District of West Bengal. Indian Ground Water xv: 7-19 ISSN 2347-8063
- ◆ Hazra S, **Roy S**, Mitra S (2017) Enhancing Adaptive Capacity and Increasing Resilience of Small and Marginal Farmers of Purulia and Bankura Districts, West Bengal to Climate Change. DRCSC Report. pp1-114.

Certificates ●

- 2018 SAC and IEEE GRSS: Soil Moisture and Agriculture monitoring using Microwave Remote Sensing.
- 2018 Centre for Groundwater Studies and Geological Society of India: Water Resources of Eastern and North-Eastern State of India.
- 2017 ESRI India: Introduction to GIS.

**Profession
Relevant
Skills**

2017 DECMA: Qualitative Research

- Strong critical and analytical skills in reading, writing and comprehension.
- Pleasant personality to connect with students and colleagues. Patience, understanding and willingness to help.
- Skilled at explaining and illustrating material in a manner that student of varying levels can understand.
- Great interest in research to advance knowledge leads to creativity, development, and success in various fields, as well as the ability to improve continuously.
- Contributed to departmental goals by serving on various committees focused on curriculum development, assessment strategies, and accreditation requirements.

Motivation

An enthusiastic, adaptive, and fast-learning person with a broad and acute interest in the discovery of new innovative ways for sustainable development. I particularly enjoy collaborating with researchers from different disciplines to develop new skills and solve new challenges.